

Math 428, Spring 2024, Homework 3 Solutions

Section 4.4: 1

Solution. $f(z) = \frac{1}{2(z - \frac{1}{2})(z - 2)}$. Simple poles at $z = \frac{1}{2}$, $z = 2$.

$$\operatorname{Res}(f, 2) = \lim_{z \rightarrow 2} \frac{z - 2}{2(z - \frac{1}{2})(z - 2)} = \frac{1}{3}, \quad \operatorname{Res}(f, \frac{1}{2}) = \lim_{z \rightarrow \frac{1}{2}} \frac{z - \frac{1}{2}}{2(z - \frac{1}{2})(z - 2)} = -\frac{1}{3}.$$

Section 4.4: 3

Solution. $f(z) = \frac{e^z}{z^2 - 1} = \frac{e^z}{(z - 1)(z + 1)}$, $\operatorname{Res}(f, 1) = \frac{e}{2}$, $\operatorname{Res}(f, -1) = -\frac{e^{-1}}{2}$,

$$\int_{|z|=2} \frac{e^z}{z^2 - 1} dz = 2\pi i \left(\frac{e}{2} - \frac{e^{-1}}{2} \right)$$

Section 4.4: 8

Solution. Contour $|z - 5| = 4$ encloses (simple) poles of $f(z) = \frac{\log z}{\sin z}$ at $z = \pi$ and $z = 2\pi$.

$$\operatorname{Res}(f, \pi) = \lim_{z \rightarrow \pi} \frac{(z - \pi) \log z}{\sin z} = \frac{\log \pi}{\cos \pi} = -\log \pi.$$

$$\operatorname{Res}(f, 2\pi) = \lim_{z \rightarrow 2\pi} \frac{(z - 2\pi) \log z}{\sin z} = \frac{\log(2\pi)}{\cos(2\pi)} = \log(2\pi).$$

$$\int_{|z-5|=4} \frac{\log z}{\sin z} dz = 2\pi i (\log(2\pi) - \log \pi) = 2\pi i \log 2.$$

Section 4.5: 1

Solution. On the set $|z| = 1$, we have $|3z^7| = 3$, and $|-z^3 + 1| \leq |z^3| + 1 \leq 2$. So we apply Rouché's Theorem with $g(z) = 3z^7$, and $f(z) = 3z^7 - z^3 + 1$ to see that $f(z)$ has 7 zeroes inside the unit disc, since $g(z)$ does.

Section 4.5: 2

Solution. On the set $|z| = 1$, we have $|-4z^3| = 4$, and $|z^5 + z - 1| \leq |z^5| + |z| + 1 \leq 3$. So we apply Rouché's Theorem with $g(z) = -4z^3$, and $f(z) = z^5 - 4z^3 + z - 1$ to see that $f(z)$ has 3 zeroes inside the unit disc, since $g(z)$ does.